MIL-C-36491A(DSA-DM)
26 January 1976
SUPERSEDING
MIL-C-36491
20 June 1966

MILITARY SPECIFICATION

CURETTES, PERIODONTAL, GRACEY

This specification is approved for use by the Defense Personnel Support Center, Department of Defense, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers double-ended periodontal curettes of Gracey design.

1.2 Classification. Curettes shall be of the following Gracey numbers, as specified (see 6.1):

Gracey	No. 1 and 2
Gracey	No. 3 and 4
Gracey	No. 5 and 6
Gracey	No. 7 and 8
Gracey	No. 9 and 10
Gracey	No. 11 and 12
Gracey	No. 13 and 14

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

FSC 6520

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116 - Methods of Preservation - packaging

STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center.

24370 - Curettes, Periodontal, Gracey.

(Copies of specifications, standards, and drawing required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

ASTM E92 - Standard Method of Test for Vickers Hardness of Metallic Materials.

ASTM B135 - Specification for Seamless Brass Tube

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General requirements. The curettes shall be in accordance with applicable requirements of GG-I-526 with the exceptions and with the additional requirements specified herein.

3.2 Standard sample. The final contour, finish and workmanship of the blade shall be equal to or better than the standard sample (see 3.6, 3.8, 3.11 and 6.3).

3.3 Bid sample. Twelve bid samples which are representative of the curettes to be supplied shall be furnished complete with protocols and satisfactory report of tests performed in accordance with the specification to the Defense Personnel Support Center for evaluation by the Military Medical Services. When tested by periodontists under conditions of normal usage, the curettes shall be suitable for use in periodontal surgical procedures. The curettes shall be well balanced and the blade shall be finished and contoured to produce a sharp edge around the blade which shall effectively accomplish subgingival curettage, remove serumal calculus from root surfaces and thoroughly polish the cementum to assist in physiological tissue reattachment.

3.4 Material.

3.4.1 Operative ends. The operative ends shall be Class 4, Type 440A, 440B, or 440C stainless steel.

3.4.2 Handle. Handle shall be chromium or nickel plated brass. Brass shall be copper alloy No. 330, as specified in ASTM B135 for chemical composition only. The corrosion tests, as specified in 3.9 shall apply.

3.5 Hardness. After proper stress relieving, the Rockwell hardness of the operative ends shall be C52 to C56 for Type 440A, C54 to C58 for Type 440B, and C56 to C60 for Type 440C. Handle and shanks shall be of proper temper and shall have a hardness suitable for the purpose intended. Hardness shall be determined as specified in 4.4.1.

3.6 Style, design and dimensions. The curettes shall be double-ended Gracey curettes. Curettes shall be furnished in the applicable number specified. The curettes shall be fine bladed and suitable for use in subgingival curettage, removal of serumal calculus from root surfaces, and for thoroughly polishing the cementum to assist in physiological tissue reattachment during periodontal treatment. The curettes shall be designed to provide push and pull action during curettage with minimum damage to tissues. The curettes shall be suitable for the purpose specified in table I. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

style, design and dimensions of the curettes shall conform to Defense Personnel Support Center (DPSC) Drawing 24370 for the applicable number. The final contour of the blade shall be equal to the standard sample (see 3.2 & 6.3).

TABLE I. Gracey numbers and purpose

Gracey Number	Purpose
1 and 2	anterior teeth
3 and 4	anterior teeth
5 and 6	bicuspids and molars
7 and 8	bicuspids and molars
9 and 10	less accessible root surfaces of molars
11 and 12	mesio-lingual root surfaces of bicuspids and molars
13 and 14	distal root surfaces of bicuspids and molars

3.6.1 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design, function, fit, assembly and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.7 Construction. The curettes shall be of three-piece construction. The operative ends shall be permanently secured to the handle. The blades of the operative ends of each curette shall face in opposite directions. Handle shall be hollow and shall have superficial diamond serrations on seven faces, and shall have one smooth face for identification marking as shown on DPSC drawing.

3.8 Finish. The operative ends including shank shall have a smooth mirror or smooth bright finish, except for the surfaces that form the cutting edges which shall have a finely honed finish. The operative ends (entire blade) shall have a sharp cutting edge and shall be free of defects such

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

as rough grind marks, cracks, gouges, burrs, feathers, or other imperfections when viewed as specified in 4.3.1. Honed portion of operative end shall be blended at junction of shank to form a smooth contour. The finish of the blades (honed surfaces) shall be equal to or better than the standard sample (3.2 and 6.3). Handle shall have a mirror or bright finish. Finger grip or serrated portions of handle shall be free from barbs and shall provide adequate finger grip during use.

3.9 Corrosion resistance. The copper sulfate test requirements of GG-I-526 shall apply to the operative ends of the curette. The boil test requirements of GG-I-526 shall apply to the entire curette.

3.10 Identification marking. In addition to the requirements of GG-I-526, each curette shall be permanently and legibly marked with the word "Gracey" and the respective Gracey instrument number, as applicable. Each instrument shall be marked on the front face. All marking shall be on the smooth portion of the handle and shall be permanent and legible. Each curette shall be marked with the name or registered trademark of the manufacturer (and contractor, if other than the manufacturer). The word "stainless," when applicable, shall also be marked on instruments of foreign manufacture. As an alternate to the word "stainless," the use of a registered trade name identifying the steel as stainless shall be acceptable, provided prior approval is received from the cognizant contracting officer.

3.11 Workmanship. In addition to the requirements of GG-I-526, the operative ends shall be free of pits, discoloration, and rounded edges. Workmanship shall be equal to or better than the standard sample (see 3.2 and 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the raw material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II, III, IV (see 4.2.1, 4.2.2 and 4.3), 4.3.1 and 4.3.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table II. The unit of product for examination purposes shall be one curette.

TABLE II. Examination criteria

Type Defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-1	2.5
Minor	II	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III except that the acceptance number shall be zero (0) for all sample sizes. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. The unit of product for test purposes shall be one curette. Lots for hardness testing shall be as specified in 4.2.2.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

TABLE III. Tests

Component	Characteristic	Reqmt.	Test Procedure	Inspection level
Entire curette	Chemical composition	3.4.1, 3.4.2, and GG-I-526	GG-I-526	S-1 1/
Operative ends	Hardness	3.5 and GG-I-526	4.4.1	S-2
Operative ends	Copper Sulfate	3.9	GG-I-526	S-2
Entire curette	Boil test	3.9	GG-I-526	S-2

1/ A sufficient quantity of each lot of metal used in the construction of the curettes may be selected for composition analysis in lieu of sample curettes, provided lot identity has been maintained and can be demonstrated to the Government.

4.2.2.1 Lots for hardness criteria. For finished instrument (operative end) hardness testing, a lot or batch of instruments shall consist of instruments (operative ends) produced by one manufacturer at one plant, from the same material (same raw material heat and heat treatment) and from the same heat treatment lot or batch (instruments undergoing heat treatment at the same time, in the same tray, rack or other containing device).

4.3 Examination. The curettes shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with 4.2.1. The curettes shall be examined for defects, including but not limited to, those listed in table IV.

TABLE IV. Classification of defects

Categories	Defects
Critical	None defined.
Major A	
101	Style and design not as specified for the applicable number curette.
102	Instrument not constructed as specified.
103	Operative end finish not as specified.
104	Operative end not free from rough grind marks, cracks, burrs, gouges, feathers, other imperfections that would render the curette unsuitable for use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

TABLE IV. Classification of defects (cont'd)

Categories	Defects
Major A	
105	Cutting edges rounded or not sharp.
106	Honed portion of operative ends not blended at junction of shank to form a smooth contour.
Major B	
151	Operative end dimensions not as specified.
Minor	
201	Handle finish not as specified.
202	Instrument not permanently and legibly marked with name or trademark of the contractor.
203	Instrument identification not present, properly located, or not complete.
204	Handle dimensions not as specified.
205	Curette not free from dirt, oil, compound, or foreign matter.
206	Handle serrations missing, not uniform or not complete.

4.3.1 Examination under magnification. The instruments shall be examined under four power (4X) magnification for the following number of defects: Categories 104 and 105. Examination under magnification shall include, but shall not be limited to, the defects listed herein.

4.3.2 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with the requirements of this specification (see 4.2.2).

4.4.1 Hardness. Hardness shall be determined in accordance with ASTM Designation E18 (Rockwell) or E92 (Vickers or Diamond Pyramid).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. Each curette of the numbers specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Curettes shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.2.2 Drying. Curettes shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservative application. Preservatives shall not be used.

5.1.2.4 Unit packaging. Each curette shall be wrapped in neutral tissue paper or other suitable non-corrosive material and packaged in accordance with MIL-P-116, method IC-2.

5.1.2.5 Intermediate package. Twelve unit packages containing curettes of one number only shall be packaged in a box constructed in accordance with PPP-B-566, PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Curettes shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Twenty-four intermediate packages (288 units) containing curettes of one number only shall be packaged in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twenty-four intermediate packages (288 units) containing curettes of one number only shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. The packaged curettes shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Each unit and intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129.

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Approval of trade name if applicable (see 3.9).

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.3 Standard sample. For access to standard sample, address the procuring activity issuing the invitation for bids or request for proposal (see 3.2, 3.6, 3.8 and 3.11).

Military Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity: DSA-DM
Project No. 6520-1417

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-36491A(DSA-DM) CURETTES, PERIODONTAL, GRACEY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, PA 19101

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S.MAIL

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:41Z
Check the source to verify that this is the current version before use.